# Chapter 7, Problem 8PP

## Problem

If

$$h(t) = \begin{cases} 0, & t < 0 \\ -4, & 0 < t < 2 \\ 3t - 8, & 2 < t < 6 \\ 0, & t > 6 \end{cases}$$

express *h*(*t*) in terms of the singularity functions.

## Step-by-step solution

**Step 1** of 4

The signal $h(t)$ is defined for $t < 0, 0 < t < 2, 2 < t < 6, t > 6$.

Represent $h(t)$ in terms of $u(t)$ for $t < 0$.

$h(t) = 0(u(-t))$ …… (1)

**Step 2** of 4

Consider $h(t)$ for $0 < t < 2$.

$h(t) = -4 \quad 0 < t < 2$

Represent $h(t)$ in terms of $u(t)$ for $0 < t < 2$.

$h(t) = -4[u(t) - u(t-2)]$ …… (2)

**Step 3** of 4

Consider $h(t)$ for $2 < t < 6$.

$h(t) = 3t - 8 \quad 2 < t < 6$

Represent $h(t)$ in terms of $u(t)$ for $2 < t < 6$.

$$\begin{aligned} h(t) &= (3t-8)[u(t-2) - u(t-6)] \\ &= 3tu(t-2) - 8u(t-2) - 3tu(t-6) + 8u(t-6) \\ &= 3(t-2+2)u(t-2) - 8u(t-2) - 3(t-6+6)u(t-6) + 8u(t-6) \\ &= 3(t-2)u(t-2) + 6u(t-2) - 8u(t-2) - 3(t-6)u(t-6) - 18u(t-6) + 8u(t-6) \\ &= 3(t-2)u(t-2) - 2u(t-2) - 3(t-6)u(t-6) - 10u(t-6) \end{aligned}$$

**Step 4** of 4

Since $r(t) = tu(t)$.

$h(t) = 3r(t-2) - 2u(t-2) - 3r(t-6) - 10u(t-6)$ …… (3)

Consider $h(t)$ for $t > 6$.

For $t > 6$, $h(t)$ is zero.

$h(t) = 0$ …… (4)

Combine equations (1), (2), (3), (4) and write equation for $h(t)$.

$$\begin{aligned} h(t) &= -4\left[u(t) - u(t-2)\right] + 3r(t-2) - 2u(t-2) - 3r(t-6) - 10u(t-6) \\ &= -4u(t) + 2u(t-2) + 3r(t-2) - 3r(t-6) - 10u(t-6) \end{aligned}$$

Therefore, $h(t)$ can be represented as

$$\boxed{h(t) = -4u(t) + 2u(t-2) + 3r(t-2) - 10u(t-6) - 3r(t-6)}.$$